

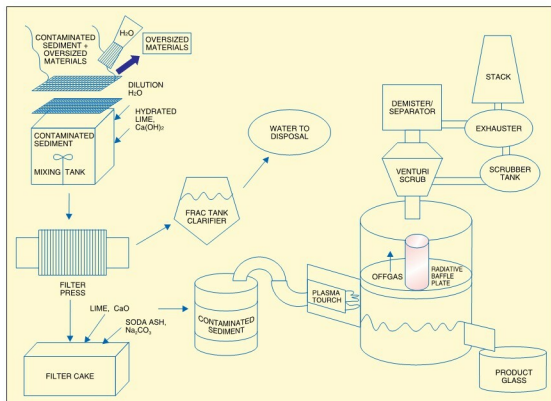


Fig. 8: Schematic of integrated plasma gasification combined cycle (IPGCC) process

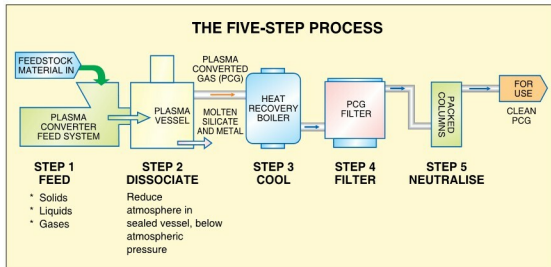


Fig. 9: Startech's plasma converter system

source at atmospheric pressure, allowing the gasifier system to be fuel-flexible, have high availability for base load power, achieve higher efficiency, and provide lower operation and maintenance costs.

The PGV reactor employs plasma torches to heat the reactor to 4000-

5000°C at atmospheric pressure. At this operating temperature, the PGV process uses a carbon-based catalyst and oxygen-enriched air to cause the hydrocarbon or organic material to undergo partial oxidation, i.e., the hydrocarbon material is depolymerised into carbon and hydrogen molecules

by the plasma-generated heat and undergoes partial oxidation to be released as a mixture of H_2 and CO . The syngas has a heating value of about one-third to one-sixth of natural gas. All inorganic or non-hydrocarbon-based materials in the feedstock are simultaneously vitrified into an inert glassy slag, suitable for use as construction materials including aggregates, tiles or bricks.

The IPGCC system uses General Electric's LCV gas turbines. The process is shown in Fig. 8.

Utilising biomass as a fuel source, Solena's technology can be used anywhere in the world. Presently, plants based on Solena's IPGCC technology are operational in North America, the Caribbean, Europe and South America.

Solena Group's ongoing projects include:

1. Five 40MW renewable energy plants in California
2. A renewable energy plant for a hazardous waste management group in Europe to turn solid and liquid waste into clean energy
3. Two 90MW renewable energy plants for two major European cities